

① Fraud Transaction Detection (Linear Search)

Java Program

```
import java.util.Scanner;
public class Main {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int[] arr = new int[N];

        for (int i = 0; i < N; i++) {
            arr[i] = sc.nextInt();
        }

        int suspicious = sc.nextInt();
        boolean found = false;

        for (int i = 0; i < N; i++) {
            if (arr[i] == suspicious) {
                found = true;
                break;
            }
        }

        if (found) {
            System.out.println("Fraud Transaction on Found");
        }
        else {
            S.o.p ("Transaction not found");
        }
    }
}
```

3

Sc.close();

2

3

```
> Main > main(String[])
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int[] transactions = new int[N];
        for (int i = 0; i < N; i++) {
            transactions[i] = sc.nextInt();
        }

        // Read the suspicious transaction to search for
        int suspicious = sc.nextInt();
        boolean found = false;

        // Perform Linear Search
        for (int i = 0; i < N; i++) {
            if (transactions[i] == suspicious) {
                found = true;
                break;
            }
        }
        if (found) {
            System.out.println(x: "Fraud Transaction Found");
        } else {
            System.out.println(x: "Transaction not Found");
        }

        sc.close();
    }
}
```


Java Program

```
import java.util.Scanner;  
public class Main {  
    public static void main (String [] args) {  
        Scanner sc = new Scanner (System.in);
```

```

int N = sc.nextInt();
int[] logs = new int[N];
for (int i = 0; i < N; i++) {
    logs[i] = sc.nextInt();
}

int target = sc.nextInt();
int low = 0, high = N - 1;
boolean found = false;
while (low <= high) {
    int mid = (low + high) / 2;
    if (logs[mid] == target) {
        found = true;
        break;
    }
    else if (logs[mid] < target) {
        low = mid + 1;
    }
    else {
        high = mid - 1;
    }
}

if (found) {
    System.out.println("log found");
}
else {
    s.o.p("log not found");
}

sc.close();
}

```



```

> Main > main(String[])
import java.util.Scanner;

public class Main {
    Scanner sc = new Scanner(System.in);
    int N = sc.nextInt();
    int[] logs = new int[N];
    for (int i = 0; i < N; i++) {
        logs[i] = sc.nextInt();
    }
    int target = sc.nextInt();

    int low = 0;
    int high = N - 1;
    boolean found = false;
    while (low <= high) {
        int mid = (low + high) / 2;

        if (logs[mid] == target) {
            found = true;
            break;
        } else if (logs[mid] < target) {
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }

    if (found) {
        System.out.println(x: "log found");
    } else {
        System.out.println(x: "log not found");
    }

    sc.close();
}

```

```

codeLabs> cd "c:\Users\Rishabh\
Desktop\decodeLabs\" ; if ($?)
{ javac Main.java } ; if ($?)
{ java Main }
7
1001
1005
1010
1015
1020
1030
1040
1015
log found
PS C:\Users\Rishabh\Desktop\de
codeLabs>

```


c Sort Employee Salaries (Bubble Sort)

```
import java.util.Scanner;  
public class Main {  
    public static void BubbleSort(int[] arr) {  
        int n = arr.length;  
        for (int i = 0; i < n - 1; i++) {  
            for (int j = 0; j < n - i - 1; j++) {  
                if (arr[j] > arr[j + 1]) {  
                    int temp = arr[j];  
                    arr[j] = arr[j + 1];  
                    arr[j + 1] = temp;  
                }  
            }  
        }  
    }  
}
```

```
public static void main(String[] args) {  
    Scanner scanner = new Scanner(System.in);  
    if (scanner.hasNextInt()) {  
        int n = scanner.nextInt();  
        int[] salaries = new int[n];  
        for (int i = 0; i < n; i++) {  
            if (scanner.hasNextInt()) {  
                salaries[i] = scanner.nextInt();  
            }  
        }  
    }  
}
```

```
    bubbleSort(salaries);
```

```
    for (int i = 0; i < n; i++)
```

```
        System.out.print(salaries[i] + (i == n - 1 ? "" : " "));
```

```
    }
```

Page No. _____

Date _____

```
System.out.println();
```

```
}
```

```
scanner.close();
```

```
}
```

```
}
```


main.java / ...

```
import java.util.Scanner;

public class Main {
    public static void bubbleSort(int[] arr) {
        int n = arr.length;

        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    // Swap elements
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Read array size
        if (scanner.hasNextInt()) {
            int n = scanner.nextInt();

            int[] salaries = new int[n];

            // Read array elements
            for (int i = 0; i < n; i++) {
                if (scanner.hasNextInt()) {
                    salaries[i] = scanner.nextInt();
                }
            }

            bubbleSort(salaries);
            for (int i = 0; i < n; i++) {
                System.out.print(salaries[i]);

                if (i < n - 1) {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
        scanner.close();
    }
}
```

Run | Debug

```
> cd "c:\Users\Rishabh\I
esktop\decodeLabs\" ; if ($?) {
    javac Main.java } ; if ($?) {
    java Main }
5
45000
32000
55000
28000
60000
28000 32000 45000 55000 60000
PS C:\Users\Rishabh\Desktop\de
codeLabs>
```

Ln 45, Col 1 Spaces: 4 LITE-8 CR LF Java

Sort website Response Time (insertion sort)

```
public class Mains  
{  
    public static void main (String[] args) {  
        Scanner sc = new Scanner (System.in);  
        int N = sc.nextInt();  
        int[] arr = new int[N];
```

```
        for (int i = 0; i < N; i++) {  
            arr[i] = sc.nextInt();  
        }
```

// insertion sort

```
        for (int i = 1; i < N; i++) {  
            int key = arr[i];  
            int j = i - 1;  
            while (j >= 0 & arr[j] > key) {  
                arr[j + 1] = arr[j];  
                j--;  
            }
```

```
            arr[j + 1] = key;  
        }
```

```
        for (int i = 0; i < N; i++) {  
            System.out.print (arr[i] + " ");  
        }
```

```
        sc.close();  
    }  
}
```

Main.java > ...

```
import java.util.Scanner;
```

```
public class Main {
```

Run | Debug

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int N = sc.nextInt();
```

```
        int[] arr = new int[N];
```

```
        for (int i = 0; i < N; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 1; i < N; i++) {
```

```
            int key = arr[i];
```

```
            int j = i - 1;
```

```
            while (j >= 0 && arr[j] > key) {
```

```
                arr[j + 1] = arr[j];
```

```
                j--;
```

```
            }
```

```
            arr[j + 1] = key;
```

```
        }
```

```
        for (int i = 0; i < N; i++) {
```

```
            System.out.print(arr[i] + " ");
```

```
        }
```

```
        System.out.println();
```

```
        sc.close();
```

```
    }
```

```
}
```

```
> cd "c:\Users\Rishabh\n\nDesktop\decodeLabs\" ; if ($?) { javac Mains.java } ; if ($?) { java Mains }
```

```
5
```

```
250
```

```
120
```

```
320
```

```
180
```

```
100
```

```
100 120 180 250 320
```

```
PS C:\Users\Rishabh\Desktop\decodeLabs>
```


prioritize Bug Reports (selection sort)

```
public class Main {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        int N = sc.nextInt();
        int[] arr = new int[N];
        for (int i = 0; i < N; i++) {
            arr[i] = sc.nextInt();
        }

        // Selection sort
        for (int i = 0; i < N - 1; i++) {
            int minIndex = i;
            for (int j = i + 1; j < N; j++) {
                if (arr[j] < arr[minIndex]) {
                    minIndex = j;
                }
            }
            int temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }
        for (int i = 0; i < N; i++) {
            System.out.print (arr[i] + " ");
        }
        sc.close();
    }
}
```

Main.java > Main

```
import java.util.Scanner;

public class Main {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int N = sc.nextInt();
        int[] arr = new int[N];
        for (int i = 0; i < N; i++) {
            arr[i] = sc.nextInt();
        }
        for (int i = 0; i < N - 1; i++) {
            int minIndex = i;
            for (int j = i + 1; j < N; j++) {
                if (arr[j] < arr[minIndex]) {
                    minIndex = j;
                }
            }

            int temp = arr[i];
            arr[i] = arr[minIndex];
            arr[minIndex] = temp;
        }
        for (int i = 0; i < N; i++) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();

        sc.close();
    }
}
```

```
> cd "C:\Users\Rishabh
\Desktop\decodeLabs\" ; if ($?) { javac Main.java } ; if ($?) { java Main }
5
9
2
8
1
5
1 2 5 8 9
PS C:\Users\Rishabh\Desktop\de
codeLabs>
```